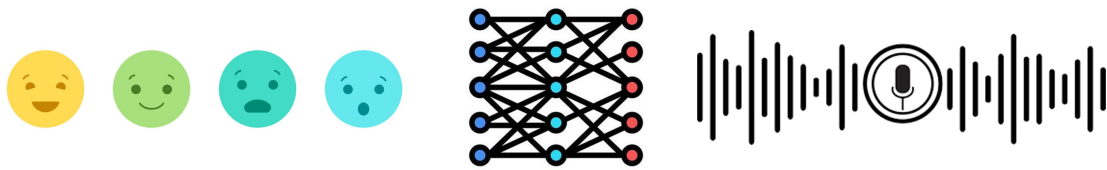


Artificial Intelligence for Digital Characters

Solution Exercise 1

Hand-out Date: 18 March 2026



Goals

- Understanding the fundamentals and methodologies of affective computing.
- Understanding the key concepts of deep learning, such as neural networks, recurrent architectures, and attention mechanisms.
- Understanding the basics, techniques, and evaluation methods of speech recognition.
- Integrating speech recognition into a Unity project to enable an avatar to understand and respond to spoken commands.

Resources

The lecture slides, exercise slides, and Unity template are accessible via Moodle.

Tasks

1. Multiple Choice Questions

a) **False.** Facial expression analysis is a non-invasive method because it involves analyzing visual data (like images or videos) without the need for physical contact. On the other hand, blood pressure measurement is invasive because it requires physical interaction, such as using a cuff or other devices to measure pressure within the arteries.

b) **False.** Action units (AUs) refer to specific movements of facial muscles, not the whole body. They are part of the Facial Action Coding System (FACS) and are used

to detect facial expressions that are linked to emotions. Whole-body movements can contribute to emotion detection, but they are not classified as action units.

c) True. ECG (Electrocardiogram) is a technique used to record the electrical activity of the heart by placing electrodes on the surface of the skin.

d) False. The softmax activation function is commonly used in classification tasks, but its output values are limited to the range $[0, 1]$, not $[-1, 1]$. The softmax function converts raw scores (logits) into probabilities by exponentiating the inputs and normalizing them. The outputs sum to 1, with each output representing the probability of a specific class.

e) False. Recurrent Neural Networks (RNNs) process time steps sequentially, one at a time, due to their inherent structure, which makes them less parallelizable compared to Transformers. RNNs maintain a hidden state that is updated at each time step, relying on information from previous time steps. In contrast, Transformers process all time steps simultaneously (in parallel) using self-attention mechanisms, making them highly parallelizable.

f) True. The attention mechanism assigns different weights to different parts of the input, allowing the model to focus on the most relevant or important information.

g) False. Analog-to-digital conversion indeed consists of two steps: sampling and quantization, but quantization does not convert real-valued numbers to positive numbers. Instead, quantization maps the continuous range of real-valued signals to a finite set of discrete values, which may include both positive and negative values depending on the signal.

h) False. Pitch is the perceptual correlate of the fundamental frequency, but the fundamental frequency is not the highest frequency component of a complex waveform. The fundamental frequency is the lowest frequency in a periodic waveform and is often the most dominant frequency.

i) False. A Mel filter bank consists of overlapping triangular filters, not rectangular filters. These triangular filters are spaced according to the Mel scale, which is designed to approximate the human ear's perception of pitch.

2. Affective Computing

a) The James-Lange theory of emotion suggests that emotions arise as a result of physiological reactions to external stimuli. The process follows this sequence:

Stimulus → Physiological Response → Emotion

For example, upon seeing a snake, the body first reacts (e.g., increased heart rate), and then the individual interprets these physiological changes as fear. Comparison with other theories:

- **Cannon-Bard Theory:** This theory proposes that physiological responses and emotions occur *simultaneously*, not sequentially. The brain processes the stimulus and triggers both emotional experience and bodily reactions at the same time.
- **Schachter-Singer Theory:** This theory suggests that emotion arises from a *combination* of physiological arousal and cognitive interpretation. The same physiological response can result in different emotions depending on how the situation is interpreted.

b)

- **Valence** measures the pleasantness of an emotion. Positive valence corresponds to happiness, while negative valence corresponds to sadness or anger.
- **Arousal** represents the intensity of an emotion, ranging from low (calm) to high (excited).
- **Dominance** indicates the sense of control one has over the emotion. High dominance means feeling in control, while low dominance means feeling overwhelmed.

Boredom is typically characterized by **Low Valence** (mildly unpleasant experience), **Low Arousal** (low energy, passive state), and **Medium-to-Low Dominance** (depends on context, but often involves a passive feeling).

c)

Step 1: Compute $f_{\text{temp}} = f - b$

$$f_{\text{temp}} = \begin{bmatrix} 120 & 130 & 20 \\ 140 & 150 & 140 \\ 110 & 150 & 120 \end{bmatrix} - \begin{bmatrix} 110 & 120 & 0 \\ 120 & 100 & 120 \\ 0 & 80 & 115 \end{bmatrix} = \begin{bmatrix} 10 & 10 & 20 \\ 20 & 50 & 20 \\ 110 & 70 & 5 \end{bmatrix}$$

Step 2: Binarizing f_{temp} **(Thresholding)**

Using a threshold $t = 5$,

$$f_{\text{binary}} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \text{ where } f_{\text{binary},i,j} = \begin{cases} 1, & \text{if } |f_{\text{temp},i,j}| > t \\ 0, & \text{otherwise} \end{cases}$$

Step 3: Compute Energy E

Energy is the percentage of surviving pixels:

$$E = \frac{\sum f_{\text{binary}}}{\text{total pixels}} \times 100 = \frac{8}{9} \times 100 = 88.89\%$$

Step 4: Update the Background $b' = (1 - \alpha)b + \alpha f$. With $\alpha = 0.2$:

$$b' = (1 - 0.2) \begin{bmatrix} 110 & 120 & 0 \\ 120 & 100 & 120 \\ 0 & 80 & 115 \end{bmatrix} + 0.2 \begin{bmatrix} 120 & 130 & 20 \\ 140 & 150 & 140 \\ 110 & 150 & 120 \end{bmatrix} = \begin{bmatrix} 112 & 122 & 4 \\ 124 & 110 & 124 \\ 22 & 94 & 116 \end{bmatrix}$$

d)

Step 1: Compute Standard Deviation (HRV Indicator)

The standard deviation σ of a sequence is given by:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}$$

where x_i are the heart rate values, $\bar{x}$ is the mean, and N is the number of values.

For $\text{seq}_1 = \{72, 75, 70, 73, 76\}$:

1. Compute the mean:

$$\bar{x}_1 = \frac{72 + 75 + 70 + 73 + 76}{5} = \frac{366}{5} = 73.2$$

2. Compute the standard deviation:

$$\sigma_1 = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2} = \sqrt{\frac{1.44 + 3.24 + 10.24 + 0.04 + 7.84}{5}} = \sqrt{\frac{22.8}{5}} \approx 2.14$$

For $\text{seq}_2 = \{74, 71, 76, 79, 80\}$:

1. Compute the mean:

$$\bar{x}_2 = \frac{74 + 71 + 76 + 79 + 80}{5} = \frac{380}{5} = 76$$

2. Compute the standard deviation:

$$\sigma_2 = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2} = \sqrt{\frac{4 + 25 + 0 + 9 + 16}{5}} = \sqrt{\frac{54}{5}} \approx 3.29$$

Step 2: Interpretation of HRV

- **Higher HRV** ($\sigma_2 = 3.29$): Greater variability in heart rate is often associated with a relaxed or adaptive emotional state. It suggests flexibility in the autonomic nervous system, linked to positive emotions and stress resilience.
- **Lower HRV** ($\sigma_1 = 2.14$): More stable heart rate is potentially linked to stress, anxiety, or reduced autonomic flexibility. Lower HRV is often associated with stress and negative emotional states.

3. Deep Learning

a) RNNs capture long-range dependencies in sequential data by maintaining a **hidden state** that is updated at each time step. This hidden state acts as a memory, allowing information to persist across multiple steps.

b) To compute the output of the neural network unit, we need to follow these steps:
1. Compute the weighted sum: The weighted sum of the inputs is given by:

$$z = \mathbf{w}^T \mathbf{x} + \mathbf{b}$$

Substituting the values:

$$z = \begin{bmatrix} 2.2 & -3 & 1.5 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \\ 8 \end{bmatrix} + 3 = (2.2 \times 0) + (-3 \times 5) + (1.5 \times 8) + 3 = 0$$

2. Apply the activation function: The activation function is the ReLU function, which outputs x if $x \geq 0$ and 0 if $x < 0$. Since $z = 0$, we have:

$$\sigma(z) = 0$$

Thus, the output of the neural network unit is 0.

c) Positional encoding is a vector added to the token embeddings that encode information about each token's position in the sequence. However, Positional encoding is **not related** to the content of the token itself. Based on the formula, the encoding for each dimension is:

$$\text{PE}(t, 1) = \cos \left(\frac{t}{10000^{0/3}} \right),$$

$$\text{PE}(t, 2) = \sin \left(\frac{t}{10000^{2/3}} \right),$$

$$\text{PE}(t, 3) = \cos \left(\frac{t}{10000^{2/3}} \right).$$

Therefore, the positional encoding for the two tokens is:

t	$\text{PE}(t, 1)$	$\text{PE}(t, 2)$	$\text{PE}(t, 3)$
1	$\cos \left(\frac{1}{10000^{0/3}} \right) \approx 0.5403$	$\sin \left(\frac{1}{10000^{2/3}} \right) \approx 0.0022$	$\cos \left(\frac{1}{10000^{2/3}} \right) \approx 1.0000$
2	$\cos \left(\frac{2}{10000^{0/3}} \right) \approx -0.4164$	$\sin \left(\frac{2}{10000^{2/3}} \right) \approx 0.0043$	$\cos \left(\frac{2}{10000^{2/3}} \right) \approx 0.9999$

d) We first compute the attention weights using the scaled dot-product similarity function:

$$\text{Attention Weight} = \frac{q^T k_i}{\sqrt{d_k}}, \quad d_k = 2$$

For pair 1:

$$q^T k_1 = 3 \times 1 + 1 \times 2 = 5, \quad \text{Attention Weight for pair 1} = \frac{5}{\sqrt{2}} \approx 3.535$$

For pair 2:

$$q^T k_2 = 3 \times 2 + 1 \times 0 = 6, \quad \text{Attention Weight for pair 2} = \frac{6}{\sqrt{2}} \approx 4.243$$

Softmax normalization:

$$\text{Softmax}(3.535, 4.243) = \left(\frac{e^{3.535}}{e^{3.535} + e^{4.243}}, \frac{e^{4.243}}{e^{3.535} + e^{4.243}} \right) \approx (0.331, 0.669)$$

Compute the output context vector:

$$\text{Context} = 0.331 \begin{bmatrix} 0 \\ 3 \end{bmatrix} + 0.669 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.669 \\ 1.662 \end{bmatrix}$$

The attention weights are 0.331 and 0.669, and the output context vector is $[0.669, 1.662]^T$.

4. Speech Recognition

a) Solution: From the figure we can see that the period T is 0.2 seconds (time for one cycle), and thus the frequency can be derived:

$$f = \frac{1}{T} = 5 \text{ Hz.}$$

For the amplitude A , it is defined as the max value on the y axis, which is $A = 2$.

b) Step 1: Convert Hertz to the Mel scale

For $f = 100 \text{ Hz}$:

$$M_{\text{low}} = 1127 \ln \left(1 + \frac{100}{700} \right) \approx 150.36 \text{ Mels.}$$

For $f = 2000 \text{ Hz}$:

$$M_{\text{high}} = 1127 \ln \left(1 + \frac{2000}{700} \right) \approx 1521.45 \text{ Mels.}$$

Step 2: Equally spaced Mel points

For 4 Mel filters, we need 6 points (including the start and end). The spacing is:

$$\Delta M = \frac{M_{\text{high}} - M_{\text{low}}}{5} \approx 274.22 \text{ Mels.}$$

The points are:

$$M_i = M_{\text{low}} + i \Delta M, \quad i = 0, 1, \dots, 5,$$

which gives:

$$\begin{aligned} M_0 &\approx 150.36, & M_1 &\approx 424.58, & M_2 &\approx 698.80, \\ M_3 &\approx 973.01, & M_4 &\approx 1247.23, & M_5 &\approx 1521.45. \end{aligned}$$

Step 3: Convert Mel points back to Hertz

The inverse transformation is:

$$f = 700 \left(e^{\frac{M}{1127}} - 1 \right).$$

Thus,

$$\begin{aligned} f_0 &= 700 \left(e^{\frac{150.36}{1127}} - 1 \right) \approx 100 \text{ Hz}, \\ f_1 &= 700 \left(e^{\frac{424.58}{1127}} - 1 \right) \approx 320 \text{ Hz}, \\ f_2 &= 700 \left(e^{\frac{698.80}{1127}} - 1 \right) \approx 600 \text{ Hz}, \\ f_3 &= 700 \left(e^{\frac{973.01}{1127}} - 1 \right) \approx 959 \text{ Hz}, \\ f_4 &= 700 \left(e^{\frac{1247.23}{1127}} - 1 \right) \approx 1415 \text{ Hz}, \\ f_5 &= 700 \left(e^{\frac{1521.45}{1127}} - 1 \right) \approx 2000 \text{ Hz}. \end{aligned}$$

Step 4: Define the Mel Filter Banks

Each triangular filter is defined by its start, peak, and end frequencies:

- **Filter 1:** start at $f_0 \approx 100$ Hz, peak at $f_1 \approx 320$ Hz, end at $f_2 \approx 600$ Hz.
- **Filter 2:** start at $f_1 \approx 320$ Hz, peak at $f_2 \approx 600$ Hz, end at $f_3 \approx 959$ Hz.
- **Filter 3:** start at $f_2 \approx 600$ Hz, peak at $f_3 \approx 959$ Hz, end at $f_4 \approx 1415$ Hz.
- **Filter 4:** start at $f_3 \approx 959$ Hz, peak at $f_4 \approx 1415$ Hz, end at $f_5 \approx 2000$ Hz.

c) To compute the WER, we use the formula:

$$\text{WER} = 100 \times \frac{\text{Insertions} + \text{Substitutions} + \text{Deletions}}{\text{Total Word in Reference}},$$

For the first case, we have two substitutions: (i) Substitution 1: “a” $\rightarrow$ “at”, and (ii) Substitution 2: “towel” $\rightarrow$ “all”. The total word in reference is 6, so we have the WER:

$$\text{WER}_1 = 100 \times \frac{2}{6} \approx 33.3\%.$$

For the second case, we have one insertion and two substitutions: (i) Substitution 1: “will” $\rightarrow$ “what”, (ii) Insertion 1: “do” is added in the output, and (iii) Substitution 2: “come” $\rightarrow$ “call”. The total word in reference is 8, so we have the WER:

$$\text{WER}_2 = 100 \times \frac{3}{8} = 37.5\%.$$